

TrES-1b

R-1575

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_200710 · created 2026-08-01T23:10:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459040.9037 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.037
Transit depth (Rp/Rs)^2	3.3e-08 +/- 0.00014 [%]
Orbital Inclination (inc)	88.89 +/- 1.49
Airmass coefficient 1 (a1)	2.12 +/- 1.01
Airmass coefficient 2 (a2)	-0.61 +/- 0.14
Scatter in the residuals of the lightcurve fit is	48.91 %
Best Comparison Star	#2 - [596, 58]
Optimal Aperture	9.75
Optimal Annulus	19.5
Transit Duration (day)	0.0886 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.037 vs 0.1358 (deviation 1.26 σ)
Duration fit vs published	0.0886 d vs 0.1042 d (deviation 0.56 σ)
Mid-time O-C	2.3 min
Depth SNR	0.0
Residual scatter	48.91 %

Light curve

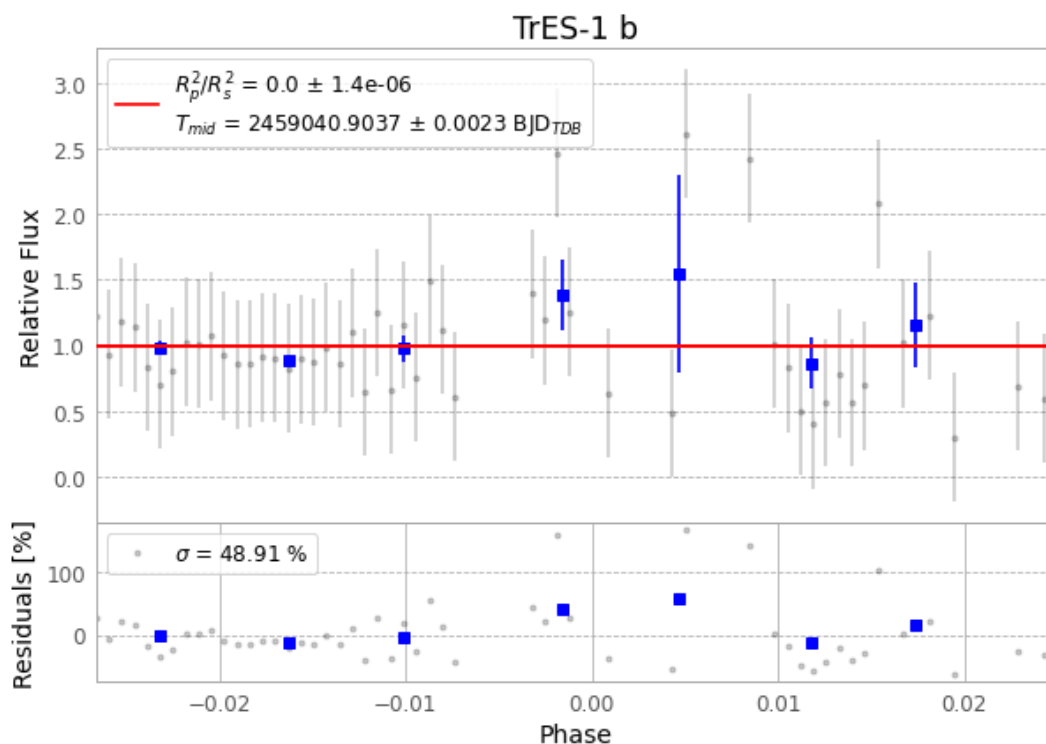


Figure 1 — FinalLightCurve_TrES-1 b_2020-07-10.png (SHA-256 dd5aeff6be890307...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [235, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 51d7f8fdd9ebeac0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), TRES-1200710073912.FITS → TRES-1200710112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-200710.log (0b2f56086ed1...), FinalLightCurve_TrES-1 b_2020-07-10.png (dd5aeff6be89...), FinalLightCurve_TrES-1 b_2020-07-10.csv (6bf980312e13...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:10Z.